

Local records (decoherence, quantum Darwinism)

system S + environment E
 $|\Psi\rangle = \sum_i c_i |s_i\rangle |e_i\rangle$

einselection: pointer states survive
 $\tau_D \sim \tau_R (\lambda_{th}/\Delta x)^2 \rightarrow 0$ for macroscopic Δx

redundancy R_δ : many fragments
 carry the same pointer record

observers $1 \dots N$ read fragments
 and agree (objectivity)

The same records summed over structure formation

collapse of bound structure; halo decoherence
 $\tau \sim \hbar R/GM^2 = 2.5 \times 10^{-87}$ s: instantaneous

encoding surface; the cosmic horizon is the largest
 IAM's Law: $k_B T \ln 2$ per bit at the nearest

accumulated ledger $E(a) = \exp(1 - 1/a)$;
 rate $\dot{I} \propto \rho_m D^n f H$ with $n = 7/2$

matter: $\mu(a) < 1$, $\mu_0 = -0.136$
 light on null geodesics: $\Sigma = 1$

the virial half sets how much bound energy writes ($\beta_m = \Omega_m/2$)